

Introduction to Calculus

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Theorem 0: The Squeeze Theorem If $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing c , except possibly at c itself, and if $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} h(x) = L$, then $\lim_{x \rightarrow c} g(x) = L$

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Theorem 1: Intermediate Value Theorem (IVT) If f is a continuous function on a closed interval $[a, b]$, and y_0 is any value between $f(a)$ and $f(b)$ (inclusive), then there is $f(c) = y_0$ at least one number c in $[a, b]$ such that

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Derivatives

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Theorem 2: Product Rule If f and g are differentiable functions, then the derivative $\frac{\partial}{\partial x}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$ of their product is

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Theorem 3: Quotient Rule If f and g are differentiable functions, and $g(x) \neq 0$, then $\frac{\partial}{\partial x}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$ the derivative of their quotient is

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